

Name : S.Dharanish

Reg no : 192472152

Couse code: CSA0925

Smart Campus Parking and Traffic Management System

1.PSEUDOCODE:

START

Connect to MySQL Database

IF connection fails

 Display "Database Connection Failed"

 STOP

END IF

Display Main Menu

REPEAT

 Display:

 1. User & Vehicle Management

 2. Parking Slot Management

 3. Reservation

 4. Entry / Exit

 5. Payments

 6. Violations

 7. Reports

 8. Exit

 Read choice

 IF choice = 1

 Add, search, update or display users

 Add and manage vehicle details

 Store data in database

 ELSE IF choice = 2

 Display parking slots

 Check slot status

 Update slot as Available, Occupied or Reserved

ELSE IF choice = 3

Read User ID, Vehicle ID and Slot ID

Check whether slot is Available

Create reservation

Change slot status to Reserved

ELSE IF choice = 4

Read Vehicle ID

Record vehicle entry

Create parking session

Record exit time

Calculate parking duration and amount

ELSE IF choice = 5

Read Session ID and Amount

Select Cash, UPI or Card

Store payment details

Set payment status

Display payment confirmation

ELSE IF choice = 6

Read User ID, Vehicle ID and Violation Type

Record description and fine amount

Store violation in database

Display violation confirmation

ELSE IF choice = 7

Calculate total users

Calculate total vehicles

Calculate available, occupied and reserved slots

Calculate reservations, payments and violations

Calculate total revenue and fines

Display reports

ELSE IF choice = 8

Close database connection

Display "Thank You"

EXIT

ELSE

Display "Invalid Choice"

END IF

UNTIL choice = 8

STOP

2. Implementation & Results

2.1 Front-End Design:

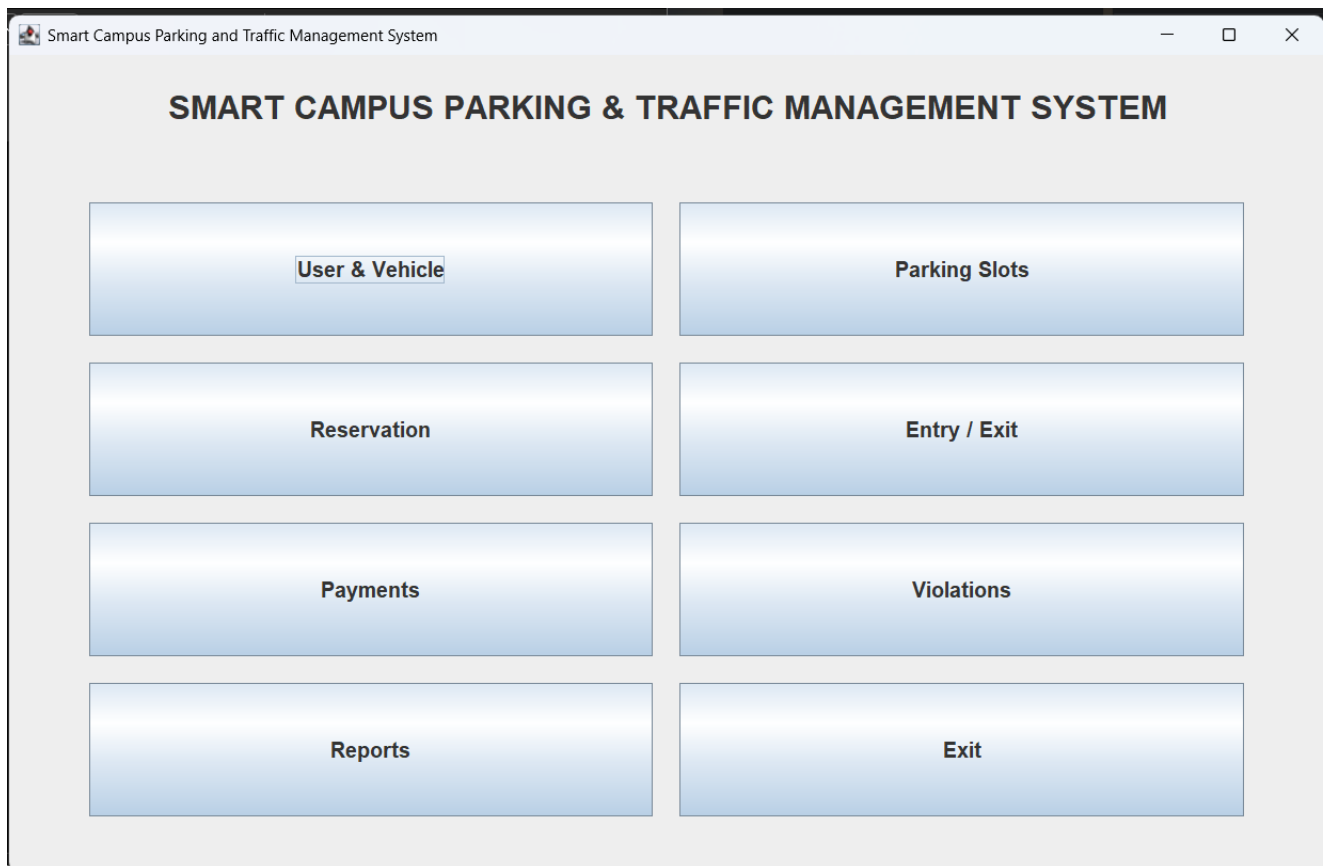


Figure 1: Main Dashboard of the Smart Campus Parking and Traffic Management System

2.2 Back-End Design:

```
mysql> SHOW TABLES;
+-----+
| Tables_in_smart_campus_parking |
+-----+
| parking_passes                  |
| parking_sessions                |
| parking_slots                   |
| parking_zones                   |
| payments                       |
| reservations                    |
| users                           |
| vehicles                       |
| violations                      |
+-----+
9 rows in set (0.07 sec)

mysql> |
```

Figure 2: MySQL Database Tables

```
mysql> DESCRIBE users;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| user_id | int | NO | PRI | NULL | auto_increment |
| name | varchar(100) | NO | | NULL | |
| email | varchar(100) | NO | UNI | NULL | |
| phone | varchar(15) | NO | | NULL | |
| user_type | enum('Student','Faculty','Staff','Visitor','Service') | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.02 sec)
```

```
mysql> DESCRIBE vehicles;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| vehicle_id | int | NO | PRI | NULL | auto_increment |
| user_id | int | NO | MUL | NULL | |
| vehicle_number | varchar(20) | NO | UNI | NULL | |
| vehicle_type | enum('Car','Bike','Van','Bus','Other') | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> DESCRIBE parking_slots;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| slot_id | int | NO | PRI | NULL | auto_increment |
| zone_id | int | NO | MUL | NULL | |
| slot_number | varchar(20) | NO | | NULL | |
| slot_status | enum('Available','Occupied','Reserved') | YES | | Available | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

```
mysql> DESCRIBE reservations;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| reservation_id | int | NO | PRI | NULL | auto_increment |
| user_id | int | NO | MUL | NULL | |
| vehicle_id | int | NO | MUL | NULL | |
| slot_id | int | NO | MUL | NULL | |
| reservation_date | date | NO | | NULL | |
| start_time | time | NO | | NULL | |
| end_time | time | NO | | NULL | |
| status | enum('Reserved','Cancelled','Completed') | YES | | Reserved | |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

```
mysql> DESCRIBE parking_sessions;
```

Field	Type	Null	Key	Default	Extra
session_id	int	NO	PRI	NULL	auto_increment
vehicle_id	int	NO	MUL	NULL	
slot_id	int	NO	MUL	NULL	
entry_time	datetime	NO		NULL	
exit_time	datetime	YES		NULL	
duration_minutes	int	YES		NULL	
parking_fee	decimal(10,2)	YES		0.00	
status	enum('Active','Completed')	YES		Active	

8 rows in set (0.00 sec)

```
mysql> DESCRIBE payments;
```

Field	Type	Null	Key	Default	Extra
payment_id	int	NO	PRI	NULL	auto_increment
session_id	int	NO	MUL	NULL	
amount	decimal(10,2)	NO		NULL	
payment_method	enum('Cash','UPI','Card')	NO		NULL	
payment_date	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
payment_status	enum('Paid','Pending')	YES		Paid	

6 rows in set (0.00 sec)

```
mysql> DESCRIBE violations;
```

Field	Type	Null	Key	Default	Extra
violation_id	int	NO	PRI	NULL	auto_increment
user_id	int	YES	MUL	NULL	
vehicle_id	int	NO	MUL	NULL	
violation_type	varchar(100)	NO		NULL	
description	varchar(255)	YES		NULL	
fine_amount	decimal(10,2)	YES		0.00	
violation_date	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
status	varchar(20)	NO		NULL	

8 rows in set (0.00 sec)

2.3 Implementation

The screenshot shows a web application titled "User & Vehicle Management". It is divided into two main sections: "User Details" and "Vehicle Details".

User Details Section:

- Fields: Name, Email, Phone, and User Type (a dropdown menu currently showing "Student").
- Buttons: "Add User" and "View Users".

Vehicle Details Section:

- Fields: User ID, Vehicle Number, and Vehicle Type (a dropdown menu currently showing "Car").
- Buttons: "Add Vehicle" and "View Vehicles".

At the bottom of the interface, there is a section labeled "Database Records".

Figure 4: User & Vehicle Management

Parking Slot Management

PARKING SLOT MANAGEMENT

View All Slots

Available Slots

Refresh

ID	ZONE	SLOT	STATUS
1	Student Zone	S01	Reserved
2	Student Zone	S02	Available
3	Student Zone	S03	Occupied
4	Student Zone	S04	Available
5	Faculty Zone	F01	Available
6	Faculty Zone	F02	Reserved
7	Faculty Zone	F03	Available
8	Visitor Zone	V01	Available
9	Visitor Zone	V02	Available

Figure 5: Parking Slot Management

Parking Reservation Management

PARKING RESERVATION MANAGEMENT

Reservation Details

User ID:

Vehicle ID:

Parking Slot:

2 - S02

Date (YYYY-MM-DD):

2026-08-31

Start Time (HH:MM):

09:00

End Time (HH:MM):

10:00

Reserve Slot

View Reservations

Refresh Slots

Figure 6: Reservation Management

Vehicle Entry and Exit Management

VEHICLE ENTRY / EXIT MANAGEMENT

Vehicle Entry

Vehicle ID:

Parking Slot:
2 - S02

Vehicle Exit

Session ID:

Record Vehicle Exit
View Parking History

Record Vehicle Entry
Refresh Slots

Figure 7: Entry / Exit Management

Parking Payment Management

PARKING PAYMENT MANAGEMENT

Payment Details

Session ID:

Amount:

Payment Method:
Cash

Payment Records

Payment ID	Session ID	Amount	Payment Method	Payment Date	Status
1	1	₹10.00	UPI	2026-08-31 22:18:05	Paid
2	1	₹10.00	Cash	2026-08-31 22:36:47	Paid

Add Payment
View Payments
Refresh
Clear
Close

Figure 8: Payment Management

Parking Violation Management

PARKING VIOLATION MANAGEMENT

Violation Details

User ID:

5

Vehicle ID:

5

Violation Type:

Description:

Fine Amount:

Status:

Pending

Add Violation

View Violations

Refresh

1	5	5	wrong parking	vehicle parking in wrong slot ₹100.00	2026-08-31 22:27:18.0	Pending
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Figure 9: Violation Management

Smart Campus Parking - Reports

SMART CAMPUS PARKING - REPORTS

Parking System Summary

Total Users: 5

Total Slots: 9

Occupied Slots: 1

Total Reservations: 1

Total Violations: 1

Total Fines: ₹100.00

Total Vehicles: 5

Available Slots: 6

Reserved Slots: 2

Total Payments: 2

Payment Revenue: ₹20.00

Detailed Report

SMART CAMPUS PARKING SYSTEM - SUMMARY

=====

Total Users : 5

Total Vehicles : 5

Total Parking Slots : 9

Available Slots : 6

Occupied Slots : 1

Reserved Slots : 2

Total Reservations : 1

Total Payments : 2

Total Violations : 1

Payment Revenue : ₹20.00

Total Fine Amount : ₹100.00

=====

Report generated successfully.

Refresh Report

User Report

Vehicle Report

Payment Report

Violation Report

Close

Figure 10: Reports

Result:

The **Smart Campus Parking and Traffic Management System** was successfully implemented using Java Swing, JDBC, and MySQL. The application successfully connects to the database and performs the required parking management operations.